power supplied to the antenna, P_t, rather than to the radiated power P_{rad}. In view of Eq. (9.27),

$$G = \xi D. \qquad \text{(dimensionless)} \tag{9.29}$$

▶ The gain accounts for ohmic losses in the antenna material, whereas the directivity does not. For a lossless antenna, $\xi = 1$, and $G = D$. ◀

9-2.5 Radiation Resistance

To a transmission line connected between a generator supplying power P_t on one end and an antenna on the other end, the antenna is merely a load with ***input impedance*** Z_{in}. If the line is lossless and properly matched to the antenna, all of P_t is transferred to the antenna. In general, Z_{in} consists of a resistive component R_{in} and a reactive component X_{in}:

$$Z_{in} = R_{in} + jX_{in}. \tag{9.30}$$

The resistive component is defined as equivalent to a resistor R_{in} that would consume an average power P_t when the amplitude of the ac current flowing through it is I_0,

$$P_t = \tfrac{1}{2} I_0^2 R_{in}. \tag{9.31}$$

Since $P_t = P_{rad} + P_{loss}$, it follows that R_{in} can be defined as the sum of a ***radiation resistance*** R_{rad} and a ***loss resistance*** R_{loss},

$$R_{in} = R_{rad} + R_{loss}, \tag{9.32}$$

with

$$P_{rad} = \tfrac{1}{2} I_0^2 R_{rad}, \tag{9.33a}$$

$$P_{loss} = \tfrac{1}{2} I_0^2 R_{loss}, \tag{9.33b}$$

where I_0 is the amplitude of the sinusoidal current exciting the antenna. As defined earlier, the ***radiation efficiency*** is the ratio of P_{rad} to P_t, or

$$\xi = \frac{P_{rad}}{P_t} = \frac{P_{rad}}{P_{rad} + P_{loss}} = \frac{R_{rad}}{R_{rad} + R_{loss}}. \tag{9.34}$$

The radiation resistance R_{rad} can be calculated by integrating the far-field power density over a sphere to obtain P_{rad} and then equating the result to Eq. (9.33a).

Example 9-3: Radiation Resistance and Efficiency of a Hertzian Dipole

A 4-cm long center-fed dipole is used as an antenna at 75 MHz. The antenna wire is made of copper and has a radius $a = 0.4$ mm. From Eqs. (7.92a) and (7.94), the loss resistance of a circular wire of length l is given by

$$R_{loss} = \frac{l}{2\pi a}\sqrt{\frac{\pi f \mu_c}{\sigma_c}}, \tag{9.35}$$

where μ_c and σ_c are the magnetic permeability and conductivity of the wire, respectively. Calculate the radiation resistance and the radiation efficiency of the dipole antenna.

Solution: At 75 MHz,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{7.5 \times 10^7} = 4 \text{ m}.$$

The length to wavelength ratio is $l/\lambda = 4 \text{ cm}/4 \text{ m} = 10^{-2}$. Hence, this is a short dipole. From Eq. (9.24),

$$P_{rad} = \frac{4\pi R^2}{D} S_{max}. \tag{9.36}$$

For the Hertzian dipole, S_{max} is given by Eq. (9.14), and from Example 9-2, we established that $D = 1.5$. Hence,

$$P_{rad} = \frac{4\pi R^2}{1.5} \times \frac{15\pi I_0^2}{R^2}\left(\frac{l}{\lambda}\right)^2 = 40\pi^2 I_0^2 \left(\frac{l}{\lambda}\right)^2. \tag{9.37}$$

Equating this result to Eq. (9.33a) and then solving for the radiation resistance R_{rad} leads to

$$R_{rad} = 80\pi^2 (l/\lambda)^2 \quad (\Omega). \quad \textbf{(short dipole)} \tag{9.38}$$

For $l/\lambda = 10^{-2}$, $R_{rad} = 0.08\ \Omega$.

Next, we determine the loss resistance R_{loss}. For copper, Appendix B gives $\mu_c \approx \mu_0 = 4\pi \times 10^{-7}$ H/m and $\sigma_c = 5.8 \times 10^7$ S/m. Hence,

$$R_{loss} = \frac{l}{2\pi a}\sqrt{\frac{\pi f \mu_c}{\sigma_c}} = \frac{4 \times 10^{-2}}{2\pi \times 4 \times 10^{-4}}\left(\frac{\pi \times 75 \times 10^6 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}\right)^{1/2} = 0.036\ \Omega.$$

Therefore, the radiation efficiency is

$$\xi = \frac{R_{rad}}{R_{rad} + R_{loss}} = \frac{0.08}{0.08 + 0.036} = 0.69.$$

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提供给天线的功率 P_t，而不是辐射功率 P_{rad}。鉴于方程。(9.27),

$$G = \xi D. \qquad \text{(dimensionless)} \tag{9.29}$$

► 增益考虑了天线材料中的欧姆损耗，而方向性则不考虑。对于无损天线，$\xi = 1$，G = D. ◄

9-2.5 耐辐射性

对于连接在一端提供功率 P_t 的发生器和另一端天线之间的传输线，天线仅仅是一个输入阻抗为 Z_{in} 的负载。如果线路是无损的并且与天线正确匹配，则所有 P_t 都会传输到天线。一般来说，Z_{in}由电阻分量R_{in}和电抗分量X_{in}组成：

$$Z_{in} = R_{in} + jX_{in}. \tag{9.30}$$

电阻组件被定义为等效于电阻器 R_{in}，当流过它的交流电流的幅度为 I_0 时，该电阻器将消耗平均功率 P_t，

$$P_t = \tfrac{1}{2} I_0^2 R_{in}. \tag{9.31}$$

由于 $P_t = P_{rad} + P_{loss}$，因此 R_{in} 可以定义为辐射电阻 R_{rad} 和损耗电阻 R_{loss} 之和，

$$R_{in} = R_{rad} + R_{loss}, \tag{9.32}$$

和

$$P_{rad} = \tfrac{1}{2} I_0^2 R_{rad}, \tag{9.33a}$$

$$P_{loss} = \tfrac{1}{2} I_0^2 R_{loss}, \tag{9.33b}$$

其中 I_0 是激励天线的正弦电流的幅度。如前所述，辐射效率是 P_{rad} 与 P_t 之比，或者

$$\xi = \frac{P_{rad}}{P_t} = \frac{P_{rad}}{P_{rad} + P_{loss}} = \frac{R_{rad}}{R_{rad} + R_{loss}}. \tag{9.34}$$

辐射电阻 R_{rad} 可以通过对球体上的远场功率密度进行积分来计算，以获得 P_{rad}，然后将结果等同于式（1）。(9.33a)。

例 9-3：赫兹偶极子的辐射电阻和效率

使用 4 厘米长的中心馈电偶极子作为 75 MHz 的天线。天线导线由铜制成，半径为= 0.4 mm。从方程式。(7.92a) 和 (7.94) 中，长度为 l 的圆形导线的损耗电阻由下式给出

$$R_{loss} = \frac{l}{2\pi a}\sqrt{\frac{\pi f \mu_c}{\sigma_c}}, \tag{9.35}$$

其中 μ_c 和 σ_c 分别是导线的磁导率和电导率。计算偶极子天线的辐射电阻和辐射效率。

解决方案：在 75 MHz 时，

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{7.5 \times 10^7} = 4 \text{ m}.$$

长度与波长之比为 $l/\lambda = 4 \text{ cm}/4 \text{ m} = 10^{-2}$。因此，这是一个短偶极子。从方程。(9.24),

$$P_{rad} = \frac{4\pi R^2}{D} S_{max}. \tag{9.36}$$

对于赫兹偶极子，S_{max} 由等式给出。(9.14)，并根据例 9-2，我们确定 D = 1.5。因此，

$$P_{rad} = \frac{4\pi R^2}{1.5} \times \frac{15\pi I_0^2}{R^2}\left(\frac{l}{\lambda}\right)^2 = 40\pi^2 I_0^2 \left(\frac{l}{\lambda}\right)^2. \tag{9.37}$$

将此结果等同于方程式。(9.33a) 然后求解辐射电阻 R_{rad} 可得出

$$R_{rad} = 80\pi^2 (l/\lambda)^2 \quad (\Omega). \quad \textbf{(short dipole)} \tag{9.38}$$

For $l/\lambda = 10^{-2}$, $R_{rad} = 0.08\ \Omega$.

接下来，我们确定损耗电阻 R_{loss}。对于铜，附录 B 给出了 $\mu_c \approx \mu_0 = 4\pi \times 10^{-7}$ H/m 和 $\sigma_c = 5.8 \times 10^7$ S/m。因此，

$$R_{loss} = \frac{l}{2\pi a}\sqrt{\frac{\pi f \mu_c}{\sigma_c}} = \frac{4 \times 10^{-2}}{2\pi \times 4 \times 10^{-4}}\left(\frac{\pi \times 75 \times 10^6 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}\right)^{1/2} = 0.036\ \Omega.$$

因此，辐射效率为

$$\xi = \frac{R_{rad}}{R_{rad} + R_{loss}} = \frac{0.08}{0.08 + 0.036} = 0.69.$$

Thus, the dipole is 69% efficient.

Concept Question 9-5: What does the pattern solid angle represent?

Concept Question 9-6: What is the magnitude of the directivity of an isotropic antenna?

Concept Question 9-7: What physical and material properties affect the radiation efficiency of a fixed-length Hertzian dipole antenna?

Exercise 9-2: An antenna has a conical radiation pattern with a normalized radiation intensity $F(\theta) = 1$ for θ between 0° and 45° and zero for θ between 45° and 180°. The pattern is independent of the azimuth angle ϕ. Find (a) the pattern solid angle and (b) the directivity.

Answer: (a) $\Omega_p = 1.84$ sr, (b) $D = 6.83$ or, equivalently, 8.3 dB. (See EM.)

Exercise 9-3: The maximum power density radiated by a short dipole at a distance of 1 km is 60 (nW/m^2). If $I_0 = 10$ A, find the radiation resistance.

Answer: $R_{rad} = 10$ mΩ. (See EM.)

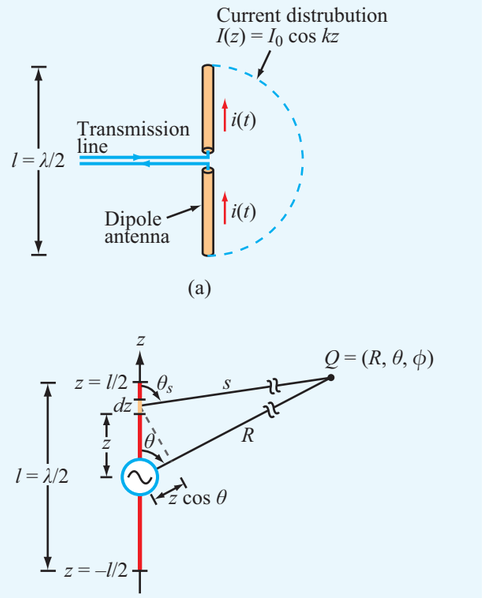


Figure 9-14 Center-fed half-wave dipole.

9-3 Half-Wave Dipole Antenna

In Section 9-1, we developed expressions for the electric and magnetic fields radiated by a Hertzian dipole of length $l \ll \lambda$. We now use these expressions as building blocks to obtain expressions for the fields radiated by a half-wave dipole antenna, so named because its length $l = \lambda/2$. As shown in Fig. 9-14, the half-wave dipole consists of a thin wire fed at its center by a generator connected to the antenna terminals via a transmission line. The current flowing through the wire has a symmetrical distribution with respect to the center of the dipole, and the current is zero at its ends. Mathematically, $i(t)$ is given by

$$i(t) = I_0 \cos \omega t \cos kz = \Re\mathrm{e}\left[I_0 \cos kz\, e^{j\omega t}\right], \qquad (9.39a)$$

whose phasor is

$$\tilde{I}(z) = I_0 \cos kz, \qquad -\lambda/4 \le z \le \lambda/4, \qquad (9.39b)$$

and $k = 2\pi/\lambda$. Equation (9.9a) gives an expression for $\tilde{E}_\theta$, which is the far field radiated by a Hertzian dipole of length l when excited by a current I_0. Let us adapt that expression to an infinitesimal dipole segment of length dz, which is excited by a current $\tilde{I}(z)$ and located at a distance s from the observation point Q (Fig. 9-14(b)). Thus,

$$d\tilde{E}_\theta(z) = \frac{jk\eta_0}{4\pi}\,\tilde{I}(z)\,dz\left(\frac{e^{-jks}}{s}\right)\sin\theta_s, \qquad (9.40a)$$

and the associated magnetic field is

$$d\tilde{H}_\phi(z) = \frac{d\tilde{E}_\theta(z)}{\eta_0}. \qquad (9.40b)$$

The far field due to radiation by the entire antenna is obtained by integrating the fields from all of the Hertzian dipoles making up the antenna:

$$\tilde{E}_\theta = \int_{z=-\lambda/4}^{\lambda/4} d\tilde{E}_\theta. \qquad (9.41)$$

Before we calculate this integral, we make the following two approximations. The first relates to the magnitude part

因此，偶极子的效率为 69%。

Concept Question 9-5: What does the pattern solid angle represent?

Concept Question 9-6: What is the magnitude of the directivity of an isotropic antenna?

Concept Question 9-7: What physical and material properties affect the radiation efficiency of a fixed-length Hertzian dipole antenna?

Exercise 9-2: An antenna has a conical radiation pattern with a normalized radiation intensity $F(\theta) = 1$ for θ between 0° and 45° and zero for θ between 45° and 180°. The pattern is independent of the azimuth angle ϕ. Find (a) the pattern solid angle and (b) the directivity.

Answer: (a) $\Omega_p = 1.84$ sr, (b) $D = 6.83$ or, equivalently, 8.3 dB. (See EM.)

Exercise 9-3: The maximum power density radiated by a short dipole at a distance of 1 km is 60 (nW/m^2). If $I_0 = 10$ A, find the radiation resistance.

Answer: $R_{rad} = 10$ mΩ. (See EM.)

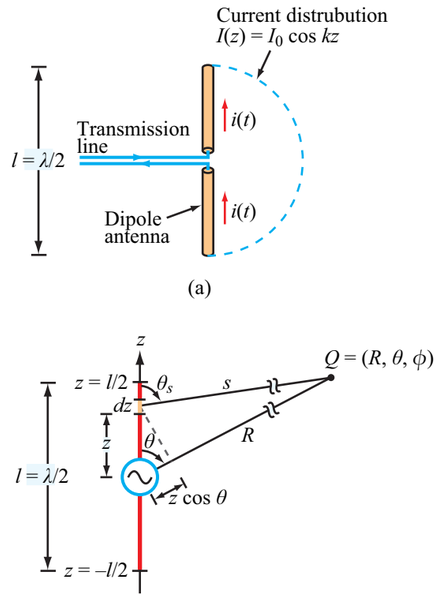


图 9-14 中心馈电半波偶极子。

9-3 半波偶极子天线

在第 9-1 节中，我们开发了长度为 l ≪ λ 的赫兹偶极子辐射的电场和磁场的表达式。现在，我们使用这些表达式作为构建块来获得半波偶极子天线辐射场的表达式，之所以如此命名是因为其长度为 l = λ/2。如图 9-14 所示，半波偶极子由一根细线组成，该细线由通过传输线连接到天线端子的发生器在其中心馈电。流过导线的电流相对于偶极子中心对称分布，两端电流为零。数学上，i(t) 由下式给出

$$i(t) = I_0 \cos \omega t \cos kz = \Re\mathrm{e}\left[I_0 \cos kz\, e^{j\omega t}\right], \qquad (9.39a)$$

谁的相量是

$$\tilde{I}(z) = I_0 \cos kz, \qquad -\lambda/4 \le z \le \lambda/4, \qquad (9.39b)$$

和 k = 2π/λ。方程（9.9a）给出了 $\tilde{E}_\theta$ 的表达式，它是长度为 l 的赫兹偶极子在受到电流 I_0 激励时辐射的远场。让我们将该表达式改编成

长度为 dz 的无穷小偶极子段，由电流 $\tilde{I}(z)$ 激发，位于距观察点 Q 距离 s 处（图 9-14（b））。因此，

$$d\tilde{E}_\theta(z) = \frac{jk\eta_0}{4\pi}\,\tilde{I}(z)\,dz\left(\frac{e^{-jks}}{s}\right)\sin\theta_s, \qquad (9.40a)$$

相关的磁场是

$$d\tilde{H}_\phi(z) = \frac{d\tilde{E}_\theta(z)}{\eta_0}. \qquad (9.40b)$$

整个天线辐射产生的远场是通过对构成天线的所有赫兹偶极子的场进行积分获得的：

$$\tilde{E}_\theta = \int_{z=-\lambda/4}^{\lambda/4} d\tilde{E}_\theta. \qquad (9.41)$$

在计算这个积分之前，我们进行以下两个近似。第一个与幅度部分有关

of the spherical propagation factor, $1/s$. In **Fig. 9-14(b)**, the distance s between the current element and the observation point Q is considered so large in comparison with the length of the dipole that the difference between s and R may be neglected in terms of its effect on $1/s$. Hence, we may set $1/s \approx 1/R$, and by the same argument, we set $\theta_s \approx \theta$. The error Δ between s and R is a maximum when the observation point is along the z axis and it is equal to $\lambda/4$ (corresponding to half of the antenna length). If $R \gg \lambda$, this error will have an insignificant effect on $1/s$. The second approximation is associated with the phase factor e^{-jks}. An error in distance Δ corresponds to an error in phase $k\Delta = (2\pi/\lambda)(\lambda/4) = \pi/2$. As a rule of thumb, a phase error greater than $\pi/8$ is considered unacceptable because it may lead to a significant error in the computed value of the field $\widetilde{E}_\theta$. Hence, the approximation $s \approx R$ is too crude for the phase factor and cannot be used. A more tolerable option is to use the ***parallel-ray approximation*** given by

$$s \approx R - z\cos\theta, \tag{9.42}$$

as illustrated in **Fig. 9-14(b)**.

Substituting Eq. (9.42) for s in the phase factor of Eq. (9.40a) and replacing s with R and θ_s with θ elsewhere in the expression, we obtain

$$d\widetilde{E}_\theta = \frac{jk\eta_0}{4\pi}\widetilde{I}(z)\,dz\left(\frac{e^{-jkR}}{R}\right)\sin\theta\, e^{jkz\cos\theta}. \tag{9.43}$$

After (1) inserting Eq. (9.43) into Eq. (9.41), (2) using the expression for $\widetilde{I}(z)$ given by Eq. (9.39b), and (3) carrying out the integration, the following expressions are obtained:

$$\widetilde{E}_\theta = j60I_0\left\{\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}\left(\frac{e^{-jkR}}{R}\right), \tag{9.44a}$$

$$\widetilde{H}_\phi = \frac{\widetilde{E}_\theta}{\eta_0}. \tag{9.44b}$$

The corresponding time-average power density is

$$S(R,\theta) = \frac{|\widetilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2}\left\{\frac{\cos^2[(\pi/2)\cos\theta]}{\sin^2\theta}\right\} = S_0\left\{\frac{\cos^2[(\pi/2)\cos\theta]}{\sin^2\theta}\right\} \quad (\mathrm{W/m^2}). \tag{9.45}$$

Examination of Eq. (9.45) reveals that $S(R,\theta)$ is maximum at $\theta = \pi/2$, and its value is

$$S_{\max} = S_0 = \frac{15I_0^2}{\pi R^2}.$$

Hence, the normalized radiation intensity is

$$F(\theta) = \frac{S(R,\theta)}{S_0} = \left\{\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}^2. \tag{9.46}$$

The radiation pattern of the half-wave dipole exhibits roughly the same doughnut-like shape shown earlier in **Fig. 9-7** for the short dipole. *Its directivity is slightly larger (1.64 compared with 1.5 for the short dipole), but its radiation resistance is 73 Ω (as shown later in Section 9-3.2), which is orders of magnitude larger than that of a short dipole.*

9-3.1 Directivity of $\lambda/2$ Dipole

To evaluate both the directivity D and the radiation resistance R_{rad} of the half-wave dipole, we first need to calculate the total radiated power P_{rad} by applying Eq. (9.20):

$$P_{\text{rad}} = R^2\iint_{4\pi} S(R,\theta)\,d\Omega = \frac{15I_0^2}{\pi}\int_0^{2\pi}\int_0^{\pi}\left\{\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}^2\sin\theta\,d\theta\,d\phi. \tag{9.47}$$

The integration over ϕ is equal to 2π, and numerical evaluation of the integration over θ gives the value 1.22. Consequently,

$$P_{\text{rad}} = 36.6I_0^2 \quad (\mathrm{W}). \tag{9.48}$$

From Eq. (9.45), we found that $S_{\max} = 15I_0^2/(\pi R^2)$. Using this in Eq. (9.24) gives the following result for the directivity D of the half-wave dipole:

$$D = \frac{4\pi R^2 S_{\max}}{P_{\text{rad}}} = \frac{4\pi R^2}{36.6I_0^2}\left(\frac{15I_0^2}{\pi R^2}\right) = 1.64 \tag{9.49}$$

or, equivalently, 2.15 dB.

9-3.2 Radiation Resistance of $\lambda/2$ Dipole

From Eq. (9.33a),

$$R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2} = \frac{2\times 36.6I_0^2}{I_0^2} \approx 73\ \Omega. \tag{9.50}$$

As was noted earlier in Example 9-3, because the radiation resistance of a Hertzian dipole is comparable in magnitude to that of its loss resistance R_{loss}, its radiation efficiency ξ is rather small. For the 4-cm long dipole of Example 9-3, $R_{\text{rad}} = 0.08\ \Omega$ (at 75 MHz) and $R_{\text{loss}} = 0.036\ \Omega$. If we keep

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球面传播因子 1/s。在图9-14(b)中，与偶极子的长度相比，电流微元和观察点Q之间的距离s被认为是如此之大，以至于s和R之间的差异对于1/s的影响可以忽略不计。因此，我们可以设置 1/s ≈ 1/R，并且通过相同的论证，我们设置 $\theta_s \approx \theta$。当观测点沿z轴时，s和R之间的误差Δ最大，等于λ/4（相当于天线长度的一半）。如果 R ≫ λ，则该误差对 1/ 的影响微不足道。第二个近似值与相位因子 e^{-jks} 相关。距离Δ的误差对应于相位 $k\Delta = (2\pi/\lambda)(\lambda/4) = \pi/2$ 的误差。根据经验，大于 π/8 的相位误差被认为是不可接受的，因为它可能会导致场 $\widetilde{E}_\theta$ 的计算值出现重大错误。因此，近似值 s ≈ R 对于相位因子来说过于粗略，无法使用。更可容忍的选择是使用由下式给出的平行射线近似

$$s \approx R - z\cos\theta, \tag{9.42}$$

如图9-14(b)所示。

代入方程式(9.42) 对于方程相位因子中的 s。 (9.40a) 并将表达式中其他地方的 s 替换为 R，将 θ_s 替换为 θ，我们得到

$$d\widetilde{E}_\theta = \frac{jk\eta_0}{4\pi}\widetilde{I}(z)\,dz\left(\frac{e^{-jkR}}{R}\right)\sin\theta\, e^{jkz\cos\theta}. \tag{9.43}$$

(1) 插入方程后： (9.43) 代入方程。 (9.41)， (2) 使用方程给出的 I(z) 表达式。 (9.39b)和(3)进行积分，得到以下表达式：

$$\widetilde{E}_\theta = j60I_0\left\{\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}\left(\frac{e^{-jkR}}{R}\right), \tag{9.44a}$$

$$\widetilde{H}_\phi = \frac{\widetilde{E}_\theta}{\eta_0}. \tag{9.44b}$$

相应的时均功率密度为

$$S(R,\theta) = \frac{|\widetilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2}\left\{\frac{\cos^2[(\pi/2)\cos\theta]}{\sin^2\theta}\right\} = S_0\left\{\frac{\cos^2[(\pi/2)\cos\theta]}{\sin^2\theta}\right\} \quad (\mathrm{W/m^2}). \tag{9.45}$$

方程的检验(9.45) 表明 S(R, θ) 在 $\theta = \pi/2$ 处最大，其值为

$$S_{\max} = S_0 = \frac{15I_0^2}{\pi R^2}.$$

因此，归一化辐射强度为

$$F(\theta) = \frac{S(R,\theta)}{S_0} = \left\{\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}^2. \tag{9.46}$$

半波偶极子的辐射图表现出与图 9-7 中短偶极子大致相同的甜甜圈形状。它的方向性稍大（1.64，而短偶极子为1.5），但其辐射电阻为73 Ω（，如后面第9-3.2)节所示，比短偶极子大几个数量级。

9-3.1 λ/2偶极子的方向性

为了评估半波偶极子的方向性 D 和辐射电阻 R_{rad}，我们首先需要通过应用方程（1）计算总辐射功率 P_{rad}。 (9.20)：

$$P_{\text{rad}} = R^2\iint_{4\pi} S(R,\theta)\,d\Omega = \frac{15I_0^2}{\pi}\int_0^{2\pi}\int_0^{\pi}\left\{\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right\}^2\sin\theta\,d\theta\,d\phi. \tag{9.47}$$

ϕ 上的积分等于 2π，并且 θ 上积分的数值计算给出值 1.22。最后，

$$P_{\text{rad}} = 36.6I_0^2 \quad (\mathrm{W}). \tag{9.48}$$

从方程。 (9.45)，我们发现 $S_{\max} = 15I_0^2/(\pi R^2)$。在等式中使用它。 (9.24) 给出半波偶极子方向性 D 的以下结果：

$$D = \frac{4\pi R^2 S_{\max}}{P_{\text{rad}}} = \frac{4\pi R^2}{36.6I_0^2}\left(\frac{15I_0^2}{\pi R^2}\right) = 1.64 \tag{9.49}$$

或等效的 2.15 dB。

9-3.2 λ/2偶极子的辐射电阻

从方程。 (9.33a)，

$$R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2} = \frac{2\times 36.6I_0^2}{I_0^2} \approx 73\ \Omega. \tag{9.50}$$

正如前面例 9-3 中所指出的，由于赫兹偶极子的辐射电阻在大小上与其损耗电阻 R_{loss} 相当，因此其辐射效率 ξ 相当小。对于示例 9-3 的 4 厘米长偶极子，75 MHz) 下的 $R_{\text{rad}} = 0.08\ \Omega$（和 $R_{\text{loss}} = 0.036\ \Omega$。如果我们保留

the frequency the same and increase the length of the dipole to 2 m ($\lambda = 4$ m at $f = 75$ MHz), R_{rad} becomes 73 Ω and R_{loss} increases to 1.8 Ω. The radiation efficiency increases from 69% for the short dipole to 98% for the half-wave dipole. More significant is the fact that it is practically impossible to match a transmission line to an antenna with a resistance on the order of 0.1 Ω, while it is quite easy to do so when $R_{rad} = 73$ Ω.

Moreover, since $R_{loss} \ll R_{rad}$ for the half-wave dipole, $R_{in} \approx R_{rad}$ and Eq. (9.30) becomes

$$Z_{in} \approx R_{rad} + jX_{in}. \tag{9.51}$$

Deriving an expression for X_{in} for the half-wave dipole is fairly complicated and beyond the scope of this book. However, it is significant to note that X_{in} is a strong function of l/λ and that it decreases from 42 Ω at $l/\lambda = 0.5$ to zero at $l/\lambda = 0.48$, whereas R_{rad} remains approximately unchanged. Hence, by reducing the length of the half-wave dipole by 4%, Z_{in} becomes purely real and equal to 73 Ω, thereby making it possible to match the dipole to a 75 Ω transmission line without resorting to the use of a matching network.

9-3.3 Quarter-Wave Monopole Antenna

▶ When placed over a conducting ground plane, a quarter-wave monopole antenna excited by a source at its base (**Fig. 9-15(a)**) exhibits the same radiation pattern in the region above the ground plane as a half-wave dipole in free space. ◀

This is because, from image theory (Section 4-11), the conducting plane can be replaced with the image of the $\lambda/4$ monopole, as illustrated in **Fig. 9-15(b)**. Thus, the $\lambda/4$ monopole radiates an electric field identical to that given by Eq. (9.44a), and its normalized radiation intensity is given by Eq. (9.46); but the radiation is limited to the upper half-space defined by $0 \le \theta \le \pi/2$. Hence, a monopole radiates only half as much power as the dipole. Consequently, for a $\lambda/4$ monopole, $P_{rad} = 18.3I_0^2$, and its radiation resistance is $R_{rad} = 36.5$ Ω.

The approach used with the quarter-wave monopole is also valid for any vertical wire antenna placed above a conducting plane, including a Hertzian monopole.

Concept Question 9-8: What is the physical length of a half-wave dipole operating at (a) 1 MHz (in the AM broadcast band), (b) 100 MHz (FM broadcast band), and (c) 10 GHz (microwave band)?

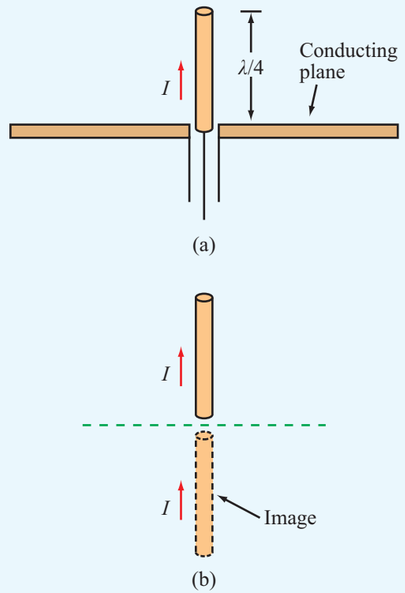


Figure 9-15 A quarter-wave monopole above a conducting plane is equivalent to a full half-wave dipole in free space.

Concept Question 9-9: How does the radiation pattern of a half-wave dipole compare with that of a Hertzian dipole? How do their directivities, radiation resistances, and radiation efficiencies compare?

Concept Question 9-10: How does the radiation efficiency of a quarter-wave monopole compare with that of a half-wave dipole, assuming that both are made of the same material and have the same cross section?

Exercise 9-4: For the half-wave dipole antenna, evaluate $F(\theta)$ versus θ to determine the half-power beamwidth in the elevation plane (the plane containing the dipole axis).

Answer: $\beta = 78°$. (See EM.)

频率相同并将偶极子长度增加到 2 m（f = 75 MHz 时 λ = 4 m），R_{rad} 变为 73 Ω，R_{loss} 增加到 1.8 Ω。辐射效率从短偶极子的 69% 增加到半波偶极子的 98%。更重要的是，实际上不可能将传输线与电阻约为 0.1 Ω 的天线相匹配，而当 $R_{rad} = 73$ Ω 时则很容易做到这一点。

此外，由于 $R_{loss} \ll R_{rad}$ 对于半波偶极子，因此 $R_{in} \approx R_{rad}$ 和等式：(9.30) 变为

$$Z_{in} \approx R_{rad} + jX_{in}. \tag{9.51}$$

推导半波偶极子 X_{in} 的表达式相当复杂，超出了本书的范围。然而，值得注意的是，X_{in} 是 l/λ 的强函数，并且它从 $l/\lambda = 0.5$ 时的 42 Ω 减少到 $l/\lambda = 0.48$ 时的零，而 R_{rad} 几乎保持不变。因此，通过将半波偶极子的长度减少4%，Z_{in}变成纯实数并且等于73Ω，从而使得可以将偶极子与75Ω传输线匹配，而无需使用匹配网络。

9-3.3 四分之一波单极天线

▶ 当放置在导电接地平面上时，由底部源激励的四分之一波单极天线（图 9-15（a））在接地平面上方的区域中表现出与自由空间中的半波偶极子相同的辐射图。 ◀

这是因为，根据镜像理论（第4-11节），传导平面可以用 λ/4单极子的镜像代替，如图9-15（b）所示。因此，λ/4 单极子辐射的电场与方程 1 给出的电场相同。 （9.44a），其归一化辐射强度由式（9.44a）给出。 （9.46）；但辐射仅限于 $0 \le \theta \le \pi/2$ 定义的上半空间。因此，单极子辐射的功率仅为偶极子的一半。因此，对于 λ/4 单极子，$P_{rad} = 18.3I_0^2$，其辐射电阻为 $R_{rad} = 36.5$ Ω。

使用四分之一波单极子的方法也适用于放置在导电平面上方的任何垂直线天线，包括赫兹单极子。

概念问题 9-8：在 (a) 1 MHz（AM 广播频段）、(b) 100 MHz（FM 广播频段）和 (c) 10 GHz（微波频段）工作的半波偶极子的物理长度是多少？

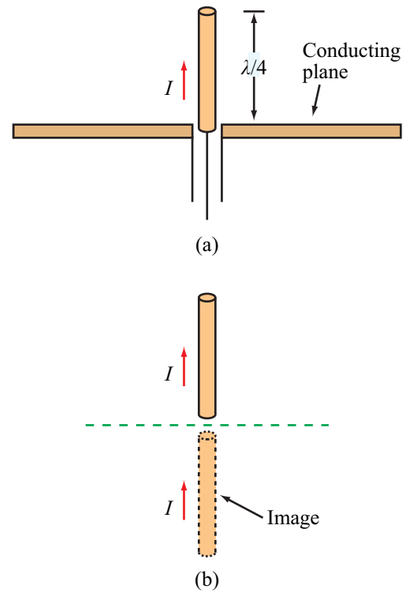


图 9-15 导电平面上方的四分之一波单极子相当于自由空间中的全半波偶极子。

概念问题 9-9：半波偶极子的辐射图与赫兹偶极子的辐射图相比如何？它们的方向性、辐射电阻和辐射效率如何比较？

概念问题 9-10：假设四分之一波长单极子和半波偶极子由相同的材料制成并且具有相同的横截面，那么它们的辐射效率如何？

练习 9-4：对于半波偶极子天线，计算 F(θ) 与 θ 的关系，以确定仰角平面（包含偶极子轴的平面）中的半功率波束宽度。

答案：β = 78°。 （参见EM。）

Exercise 9-5: If the maximum power density radiated by a half-wave dipole is 50 μW/m^2 at a range of 1 km, what is the current amplitude I_0?

Answer: $I_0 = 3.24$ A. (See EM.)

9-4 Dipole of Arbitrary Length

So far, we examined the radiation properties of the Hertzian and half-wave dipoles. We now consider the more general case of a linear dipole of arbitrary length l relative to λ. For a center-fed dipole, as depicted in **Fig. 9-16**, the currents flowing through its two halves are symmetrical and must go to zero at its ends. Hence, the current phasor $\widetilde{I}(z)$ can be expressed as a sine function with an argument that goes to zero at $z = \pm l/2$:

$$\widetilde{I}(z) = \begin{cases} I_0 \sin[k(l/2 - z)], & \text{for } 0 \le z \le l/2, \\ I_0 \sin[k(l/2 + z)], & \text{for } -l/2 \le z < 0, \end{cases} \tag{9.52}$$

where I_0 is the current amplitude. The procedure for calculating the electric and magnetic fields and the associated power density of the wave radiated by such an antenna is basically the same as that used previously in connection with the half-wave dipole antenna. The only difference is the current distribution $\widetilde{I}(z)$. If we insert the expression for $\widetilde{I}(z)$ given by Eq. (9.52) into Eq. (9.43), we obtain the following expression for the differential electric field $d\widetilde{E}_\theta$ of the wave radiated by an elemental length dz at location z along the dipole:

$$d\widetilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi}\left(\frac{e^{-jkR}}{R}\right)\sin\theta\, e^{jkz\cos\theta}\, dz \times \begin{cases} \sin[k(l/2 - z)] & \text{for } 0 \le z \le l/2, \\ \sin[k(l/2 + z)] & \text{for } -l/2 \le z < 0. \end{cases} \tag{9.53}$$

The total field radiated by the dipole is

$$\widetilde{E}_\theta = \int_{-l/2}^{l/2} d\widetilde{E}_\theta = \int_0^{l/2} d\widetilde{E}_\theta + \int_{-l/2}^{0} d\widetilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi}\left(\frac{e^{-jkR}}{R}\right)\sin\theta \times \left\{\int_0^{l/2} e^{jkz\cos\theta}\sin[k(l/2 - z)]\, dz + \int_{-l/2}^{0} e^{jkz\cos\theta}\sin[k(l/2 + z)]\, dz\right\}. \tag{9.54}$$

If we apply Euler's identity to express $e^{jkz\cos\theta}$ as

$$e^{jkz\cos\theta} = \cos(kz\cos\theta) + j\sin(kz\cos\theta)],$$

we can integrate the two integrals and obtain the result

$$\widetilde{E}_\theta = j60I_0\left(\frac{e^{-jkR}}{R}\right)\left[\frac{\cos\left(\frac{kl}{2}\cos\theta\right) - \cos\left(\frac{kl}{2}\right)}{\sin\theta}\right]. \tag{9.55}$$

The corresponding time-average power density radiated by the dipole antenna is given by

$$S(\theta) = \frac{|\widetilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2, \tag{9.56}$$

where we have used the relations $\eta_0 \approx 120\pi$ (Ω) and $k = 2\pi/\lambda$. For $l = \lambda/2$, Eq. (9.56) reduces to the expression given by Eq. (9.45) for the half-wave dipole. Plots of the normalized radiation intensity, $F(\theta) = S(R, \theta)/S_{\max}$, are shown in **Fig. 9-17** for dipoles of lengths $\lambda/2$, λ, and $3\lambda/2$. The dipoles with $l = \lambda/2$ and $l = \lambda$ have similar radiation patterns with both maxima along $\theta = 90°$, but the half-power beamwidth of the wavelength-long dipole is narrower than that of the half-wave dipole, and $S_{\max} = 60I_0^2/(\pi R^2)$ for the wavelength-long dipole, which is four times that for the half-wave dipole. The pattern of the dipole with length $l = 3\lambda/2$ exhibits a structure with multiple lobes, and its direction of maximum radiation is not along $\theta = 90°$.

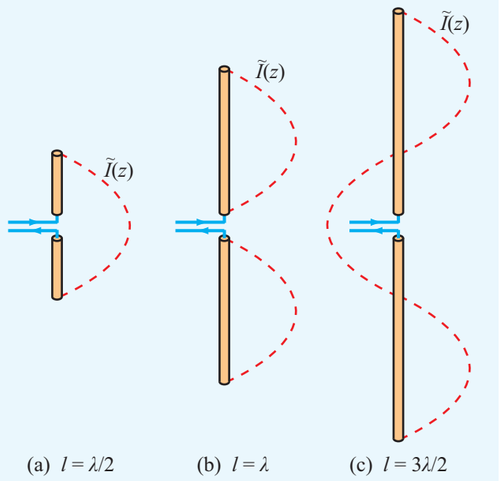


Figure 9-16 Current distribution for three center-fed dipoles.

练习9-5：如果半波偶极子在1公里范围内辐射的最大功率密度为50 μ W/m^2，则电流幅度I_0是多少？

答案：$I_0 = 3.24$ A。（参见EM。）

9-4 任意长度的偶极子

到目前为止，我们研究了赫兹偶极子和半波偶极子的辐射特性。现在我们考虑更一般的情况，即相对于 λ 具有任意长度 l 的线性偶极子。对于中心馈电偶极子，如图 9-16 所示，流过其两半的电流是对称的，并且在其两端必须为零。因此，电流相量 ~I(z) 可以表示为一个正弦函数，其参数在 z = ±l/2 处为零：

$$\widetilde{I}(z) = \begin{cases} I_0 \sin[k(l/2 - z)], & \text{for } 0 \le z \le l/2, \\ I_0 \sin[k(l/2 + z)], & \text{for } -l/2 \le z < 0, \end{cases} \tag{9.52}$$

其中 I_0 是电流幅度。计算电场和磁场以及由这种天线辐射的波的相关功率密度的过程基本上与先前用于半波偶极子天线的过程相同。唯一的区别是电流分布 ~I(z)。如果我们插入由下式给出的 ~I(z) 表达式等式。(9.52) 代入方程。根据方程 (9.43)，我们得到了沿偶极子位置 z 处的元素长度 dz 辐射的波的微分电场 d ~E θ 的表达式：

$$d\widetilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi}\left(\frac{e^{-jkR}}{R}\right)\sin\theta\, e^{jkz\cos\theta}\, dz \times \begin{cases} \sin[k(l/2 - z)] & \text{for } 0 \le z \le l/2, \\ \sin[k(l/2 + z)] & \text{for } -l/2 \le z < 0. \end{cases} \tag{9.53}$$

偶极子辐射的总场为

$$\widetilde{E}_\theta = \int_{-l/2}^{l/2} d\widetilde{E}_\theta = \int_0^{l/2} d\widetilde{E}_\theta + \int_{-l/2}^{0} d\widetilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi}\left(\frac{e^{-jkR}}{R}\right)\sin\theta \times \left\{\int_0^{l/2} e^{jkz\cos\theta}\sin[k(l/2 - z)]\, dz + \int_{-l/2}^{0} e^{jkz\cos\theta}\sin[k(l/2 + z)]\, dz\right\}. \tag{9.54}$$

如果我们应用欧拉恒等式将 e $^{jkz\cos\theta}$ 表示为

$$e^{jkz\cos\theta} = \cos(kz\cos\theta) + j\sin(kz\cos\theta)],$$

我们可以将两个积分积分并得到结果

$$\widetilde{E}_\theta = j60I_0\left(\frac{e^{-jkR}}{R}\right)\left[\frac{\cos\left(\frac{kl}{2}\cos\theta\right) - \cos\left(\frac{kl}{2}\right)}{\sin\theta}\right]. \tag{9.55}$$

偶极子天线辐射的相应时间平均功率密度由下式给出

$$S(\theta) = \frac{|\widetilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2, \tag{9.56}$$

其中我们使用了关系 $\eta_0 \approx 120\pi$ (Ω) 和 $k = 2\pi/\lambda$。对于 $l = \lambda/2$，方程。 (9.56) 简化为方程给出的表达式。(9.45) 为半波偶极子。图 9-17 显示了长度为 λ/2、λ 和 3λ/2 偶极子的归一化辐射强度 F(θ) = S(R, θ)/S_{max} 的图。具有 $l = \lambda/2$ 和 $l = \lambda$ 的偶极子具有类似的辐射方向图，最大值均沿 $\theta = 90°$，但长波长偶极子的半功率波束宽度比半波偶极子窄，并且长波长偶极子的 $S_{max} = 60I_0^2/(\pi R^2)$ 是半波偶极子的四倍。长度为$l = 3\lambda/2$ 的偶极子的图案呈现出多瓣结构，其最大辐射方向不是沿着$\theta = 90°$。

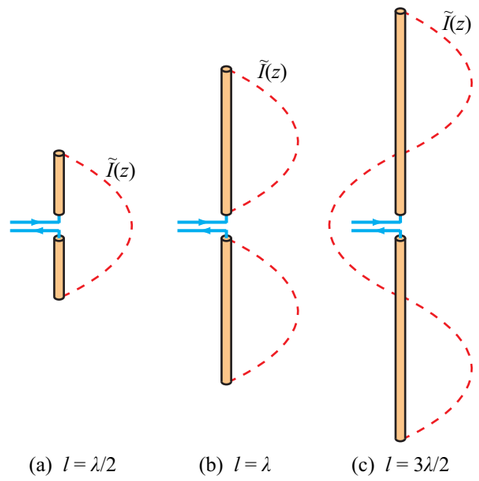


图 9-16 三个中心馈电偶极子的电流分布。